

Assignment 2 – Permissions & Ownership

1. Create a file named secure.txt

2. Set permissions so that:

○ Owner → read/write

○ Group → read only

○ Others → no access

```
ubuntu@ip-172-31-31-42:~$ pwd
/home/ubuntu
ubuntu@ip-172-31-31-42:~$ cd linux_practice
ubuntu@ip-172-31-31-42:~/linux_practice$ cd day1
ubuntu@ip-172-31-31-42:~/linux_practice/day1$ ls
backup  files
ubuntu@ip-172-31-31-42:~/linux_practice/day1$ cd files
ubuntu@ip-172-31-31-42:~/linux_practice/day1/files$ touch secure.txt
ubuntu@ip-172-31-31-42:~/linux_practice/day1/files$ ls
abc.txt  def.txt  secure.txt  xyz.txt
ubuntu@ip-172-31-31-42:~/linux_practice/day1/files$ chmod 640 secure.txt
ubuntu@ip-172-31-31-42:~/linux_practice/day1/files$ ls -l secure.txt
-rw-r----- 1 ubuntu ubuntu 0 Jan 13 11:03 secure.txt
ubuntu@ip-172-31-31-42:~/linux_practice/day1/files$
```

3. Change ownership of the file to another user.

```
ubuntu@ip-172-31-31-42:~$ sudo adduser deepika
info: Adding user `deepika' ...
info: Selecting UID/GID from range 1000 to 59999 ...
info: Adding new group `deepika' (1001) ...
info: Adding new user `deepika' (1001) with group `deepika (1001)' ...
info: Creating home directory `/home/deepika' ...
info: Copying files from `/etc/skel' ...
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for deepika
Enter the new value, or press ENTER for the default
    Full Name []: Deepika
    Room Number []:
    Work Phone []:
    Home Phone []:
    Other []:
Is the information correct? [Y/n] Y
info: Adding new user `deepika' to supplemental / extra groups `users' ...
info: Adding user `deepika' to group `users' ...
ubuntu@ip-172-31-31-42:~$ cd /home
ubuntu@ip-172-31-31-42:/home$ ls
deepika  ubuntu
```

```
ubuntu@ip-172-31-31-42:/home$ sudo chown deepika /home/ubuntu/linux_practice/day1/files/secure.txt
ubuntu@ip-172-31-31-42:/home$ ls -l /home/ubuntu/linux_practice/day1/files/secure.txt
-rw-r----- 1 deepika ubuntu 0 Jan 13 11:03 /home/ubuntu/linux_practice/day1/files/secure.txt
ubuntu@ip-172-31-31-42:/home$
```

4. Explain:

○ Why 777 is dangerous

○ Difference between permission and ownership

1. 777 means:

- Read (r), Write (w), Execute (x) for everyone:
 - Owner: full access
 - Group: full access
 - Others: full access

This is risky because any user can read, write and execute making it risky. Creates a security vulnerability by enabling modify, delete and execute access to every user.

2.

Permission:

- Defines actions like read, write, execute that are allowed for Owner, Group, Others
- Controlled using chmod

Ownership:

- Ownership determines who controls the file.
- We can change ownership using chown.